NeuroSym-BD: Neuro-Symbolic Violence Analysis from Live News Portals

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Anonymized for Double-Blind Review

1. The AI Dilemma in Automated Conflict Monitoring

Automated conflict event monitoring from live press streams faces a fundamental trade-off:

- **Unconstrained Neural Sequence Models:** Highly flexible for journalistic syntax, but susceptible to rhetorical false alarms on political hyperbole (e.g., "vow a bloodbath" miscoded as armed clashes; Zero-Shot NLI FPR = 31.3%).
- **Pure Symbolic Rule Engines:** High recall (90.2%) and deterministic, but catastrophic over-triggering on journalistic prose (86.9% FPR on non-violent cases, yielding poor precision 34.8%).

NeuroSym-BD balances this trade-off as a high-precision, provenance-grounded filter: fusing neural representations with OWL 2 DL ($\mathcal{SROIQ}$) constraint gates in BlazeGraph to suppress false alarms ($P = 96.8\%$, FPR = 1.0%) while lifting recall (45.1% → 58.8%).

2. Bangladesh CAMEO Domain Extension

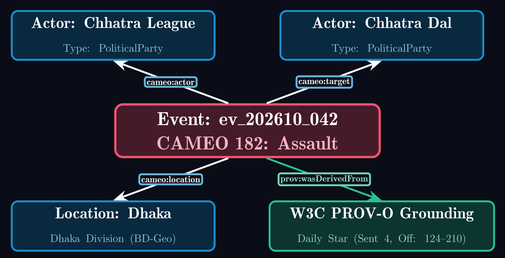
Standard CAMEO (Codes 13–20) omits localized South Asian conflict archetypes. We formulated 21 specialized OWL 2 Turtle subclasses across 6 regional macro-categories grounded in Bengali socio-political realities:

CAMEO Subcode	Macro-Category	Bangla Socio-Political Context
`cameo-bd:PV01`	Political Violence	Hartal strike violence (picket arson, transport blockades)
`cameo-bd:PV02`	Political Violence	Inter-party clash (Awami League vs BNP student wings)
`cameo-bd:SV01`	State Violence	Extrajudicial encounter (*Crossfire* / staged gunfight)
`cameo-bd:CV01`	Communal Violence	Religious communal violence (temple and minority attacks)
`cameo-bd:LV01`	Labor Violence	Garment worker unrest (RMG wage highway blockades)
`cameo-bd:OC03`	Organized Crime	Mob justice or lynching (*Gonopituni* vigilante attacks)

Every extracted event is permanently bound to W3C PROV-O provenance metadata, ensuring 98.0% verbatim sentence grounding and 86.7% trigger faithfulness.

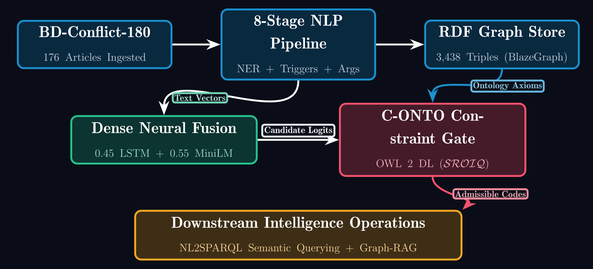
3. BlazeGraph RDF Subgraph & Provenance Grounding

Live Graph Representation with W3C PROV-O Grounding:



- **98.0% Metadata Completeness**: Verified news URI provenance links (98/100).
- **86.7% Category Faithfulness**: Explicit triggers verified in source sentence.

4. Neuro-Symbolic System Architecture



The 5 Constraint Gates:

- **C-ONTO (Hard Gate):** Enforces transition plausibility matrix via live SPARQL. Invalid jumps fallback via constrained decoding: $c^* = \arg\max_{c \in \mathcal{C}_{\text{valid}}(\text{prev})} P_{\text{neural}}(c)$ (c_{prev}: preceding event in same district timeline).
- **C-ACTOR (Soft Gate):** Enforces actor agency (e.g. non-state actors asserting artillery predicates penalize confidence $\Delta\tau = -0.05$).
- **C-GEO (Soft Gate):** Enforces 8-Division / 64-District administrative hierarchy.
- **C-TEMP / C-CONF (Soft Gates):** Chronology checks & confidence pruning ($\tau \geq 0.60$).

5. Ontological State Transition Plausibility Matrix

OWL 2 DL ($\mathcal{SROIQ}$) Transition Prior Matrix (`ontology/reasoning_rules.ttl`):

Precursor \ Target	13	14	15	16	17	18	19	20
13 (Threaten)	✓	✓	✓	✓	✓	×	×	×
14 (Protest)	✓	✓	✓	✓	✓	×	×	×
15 (Exhibit Force)	×	✓	✓	✓	✓	✓	×	×
16 (Reduce Relations)	×	×	✓	✓	✓	✓	×	×
17 (Coerce)	×	×	×	✓	✓	✓	✓	×
18 (Assault)	×	×	×	×	✓	✓	✓	✓
19 (Fight)	×	×	×	×	×	✓	✓	✓
20 (Mass Violence)	×	×	×	×	×	×	✓	✓

Formal DL Verification: OWL 2 property disjointness (valid $\sqcap$ illegal $\equiv \bot$) verified via `dl_reasoner_check.py` with **0 consistency violations**. Prohibits unescalated rhetorical jumps (13 → 20) while maintaining ongoing violence continuity (19 → 18/19).

6. Downstream Applications: NL2SPARQL & Graph-RAG

- **NL2SPARQL Benchmark ($n = 50$ Gold Queries):** 100% syntax validity, 100% schema compliance, and **52.0% execution accuracy** (district filtering 80%, provenance 100%; complex temporal/spatial windows require human review).
- **Multi-Relational KG Embeddings (PyKEEN):** Evaluated on `BD-Conflict-180` KG (654 core relational triples as low-resource NeSy benchmark); RotatE achieves **MRR = 0.369** and **Hits@10 = 0.641**, outperforming TransE (0.211).

7. Empirical Benchmark Alignment

External Benchmark	Corpus Size	Evaluated	Code Concord	Macro Concord
vs ACLED (2020–2026)	6,691 events	42 matched	**78.6%**	**100.0%** ($HV \geq 18$)
vs GDELT 2.0 (Exact URLs)	41,554 stories	78 stories	**61.5%**	**100.0%** ($HV \geq 18$)

*Non-circular GDELT protocol matches strictly on identical source URLs without CAMEO pre-filtering.

8. Controlled Ablation Study (`BD-Conflict-180` Held-Out Test Set, $n = 150$)

Model Architecture	BinAcc	Prec@HV	Rec@HV	F1@HV	Std-FPR	Provenance
Keyword Heuristic	0.453	0.363	0.804	0.500	0.727	100.0%
Pure Symbolic Rule Engine	0.393	0.348	**0.902**	0.503	0.869	100.0%
Zero-Shot NLI (DistilBERT)	0.620	0.446	0.490	0.467	0.313	0.0%
Zero-Shot NLI + Symbolic	**0.780**	**0.737**	0.549	**0.629**	**0.101**	**100.0%**
Dense Neural (MiniLM+LSTM)	0.787	0.852	0.451	0.590	0.040	0.0%
NeuroSym-BD Full Hybrid	**0.853**	**0.968**	0.588	**0.732**	**0.010**	**100.0%**

- **Model-Agnostic Gating**: C-ONTO elevates Zero-Shot NLI (F_1 : 0.467 → 0.629, FPR: 31.3% → 10.1%).
- **Significant Synergy**: Hybrid significantly improves paired classification accuracy (12 wins vs 2 losses, $p = 0.0129$, exact McNemar test), while slashing false alarms by 75% ($FP : 4 \rightarrow 1$).
- **Zero-Leakage Article Holdout ($n = 164$)**: Directionally consistent across 49 unseen articles (84.1% vs 81.7% Acc, 0.683 vs 0.625 F1; $p = 0.289$, exact binomial test, $b = 6, c = 2$).

9. Real-World Conflict Case Walkthrough

Scenario: Highway Garment Strike (Savar, Dhaka):
"Hundreds of readymade garment workers blocked the Dhaka-Aricha highway demanding unpaid wages."

- **Neural Classifier Mistake:** High lexical tension causes dense text scorer to trigger CAMEO 182 (*Physical Assault*), predicting severe armed violence (FP).
- **Symbolic C-ONTO Interception:** Gating engine scans predicate arguments: no physical trauma or assault verbs detected; precursor was labor protest (14).
- **Corrective Fallback:** Intercepts invalid jump and maps to CAMEO 144 (*Block passage* / non-violent demonstration), successfully preventing false alarm.
- **Auditable Provenance:** Generated RDF triple grounds inference directly to Savar jurisdiction with source sentence offset and 100% W3C PROV-O trace.

10. Reproducibility & Codebase

- **Complete Open-Source Stack**: Code, `BD-Conflict-180` curated silver benchmark, 21 BD-CAMEO ontology subclasses, SPARQL endpoints, and evaluation scripts released under MIT License.
- **Verification**: 28/28 unit tests pass (`make test`).

`https://github.com/anonymised/neurosym-bd`